

CREAL REEF, Australia

WMO: 943710

Lat: 20.530S

Lon: 150.377E

Elev: 6

StdP: 101.25

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 18.2	(c) 19.1	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 14.5	(k) 21.0	(l) 14.0	(m) 21.2	(n) 8.8	(o) 170	(p) 0.613

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 2	(b) 1.4	(c) 28.9	(d) N/A	(e) 28.6	(f) N/A	(g) 28.3	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 5.5	(p) 0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 14.6	(b) 13.4	(c) 12.5	(e) 16.8	(f) 29.9	(g) 0.7	(h) 0.9	(i) 16.3	(j) 30.6	(k) 15.8	(l) 31.1	(m) 15.4	(n) 31.6	(o) 14.9	(p) 32.3		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.4	27.0	27.3	26.8	25.6	23.8	22.2	21.3	21.5	22.5	23.9	25.1	26.4		
	DBStd	2.31	0.82	0.80	0.80	0.81	1.01	1.20	1.09	1.07	0.94	0.86	0.77	0.84		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	5267	527	486	521	468	428	366	350	356	376	429	453	508		
	CDD18.3	2225	268	252	262	218	170	116	92	97	126	171	203	249		
	CDH23.3	14906	2808	2761	2681	1724	594	79	7	11	81	520	1323	2318		
	CDH26.7	1612	457	582	360	34	0	0	0	0	0	0	7	172		
(14) Wind	WSAvg	7.4	7.4	7.5	8.5	8.6	8.0	8.0	7.5	6.7	6.3	6.6	6.7	6.9		
(15) Precipitation	PrecAvg	840	126	148	134	72	53	50	34	32	20	30	43	99		
	PrecMax	1503	331	520	521	218	236	147	178	167	111	103	231	347		
	PrecMin	375	28	20	12	8	11	10	3	1	1	2	6	17		
	PrecStd	250	74	101	124	54	41	34	38	39	23	25	54	75		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.2	29.7	29.2	27.9	26.2	24.9	24.0	24.2	25.4	26.2	27.4	28.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	28.7	29.1	28.7	27.4	25.7	24.5	23.5	23.6	24.7	25.7	26.8	28.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	28.5	28.7	28.3	27.1	25.5	24.1	23.1	23.2	24.2	25.4	26.5	27.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	28.2	28.4	28.0	26.7	25.1	23.8	22.8	22.8	23.7	25.1	26.2	27.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		1.4	1.4	1.4	1.3	1.5	1.5	1.5	1.3	1.2	1.1	1.2	1.3		
	5% DB	MCDBR	1.6	1.7	1.7	1.5	1.7	1.6	1.6	1.5	1.4	1.4	1.5	1.5		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.396	0.395	0.390	0.363	0.345	0.338	0.325	0.328	0.351	0.377	0.376	0.386		
	taud		2.563	2.578	2.575	2.636	2.644	2.655	2.671	2.652	2.565	2.516	2.558	2.567		
	Ebn at Noon		949	938	919	906	886	874	899	928	942	944	961	959		
	Edh at Noon		109	105	102	89	82	78	79	86	101	111	109	108		
(44) All-Sky Solar Radiation	RadAvg		6.81	6.32	5.74	5.13	4.37	3.85	4.28	5.19	6.33	7.01	7.29	7.26		
	RadStd		0.44	0.66	0.53	0.37	0.27	0.38	0.29	0.35	0.34	0.34	0.68	0.69		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page